

EXPERIMENT 4

Student Name: Vishesh Aggarwal
Branch: AIT-CSE (Core)
Semester: 5
Subject Name: Computer Network

UID: 24BCY70145
Section/Group: 24AIT_KRG-G2
Subject Code: 24CSH-337

Experiment

Implementation of Static Routing using multiple routers in Cisco Packet Tracer to establish communication between different networks.

Aim

To implement Static Routing using routers in Cisco Packet Tracer and verify communication between different networks.

Objective

- To understand the concept of routing.
 - To study the working of static routing.
 - To configure static routes manually on routers.
 - To establish communication between different networks.
 - To verify packet transmission using the Ping command.
-

Software Requirements

Software

- Cisco Packet Tracer

Hardware (Virtual)

- Routers
- Switches
- PCs

EXPERIMENT 4

- Copper Straight-Through Cables
 - Serial Cables (if used)
-

Experiment Theory

Routing is the process of finding the best path and forwarding data packets from a source network to a destination network using routers. Static routing is a routing technique in which all routes are manually configured by the network administrator. It is simple, secure, and suitable for small networks, but it does not automatically adapt to network changes.

Practical / Experiment Steps

1. Open Cisco Packet Tracer.
 2. Place the required routers, switches, and PCs.
 3. Connect all devices using appropriate cables.
 4. Configure IP addresses for all PCs and router interfaces.
 5. Configure static routes on each router using the ip route command.
 6. Save the router configuration.
 7. Test communication between different networks using the Ping command.
 8. Verify successful packet transmission.
-

Procedure

1. Design the required network topology.
2. Configure IP addresses on all end devices.
3. Configure router interfaces.
4. Add static routes manually on every router.
5. Verify routing tables.
6. Perform Ping tests between different networks.
7. Observe successful communication.

EXPERIMENT 4

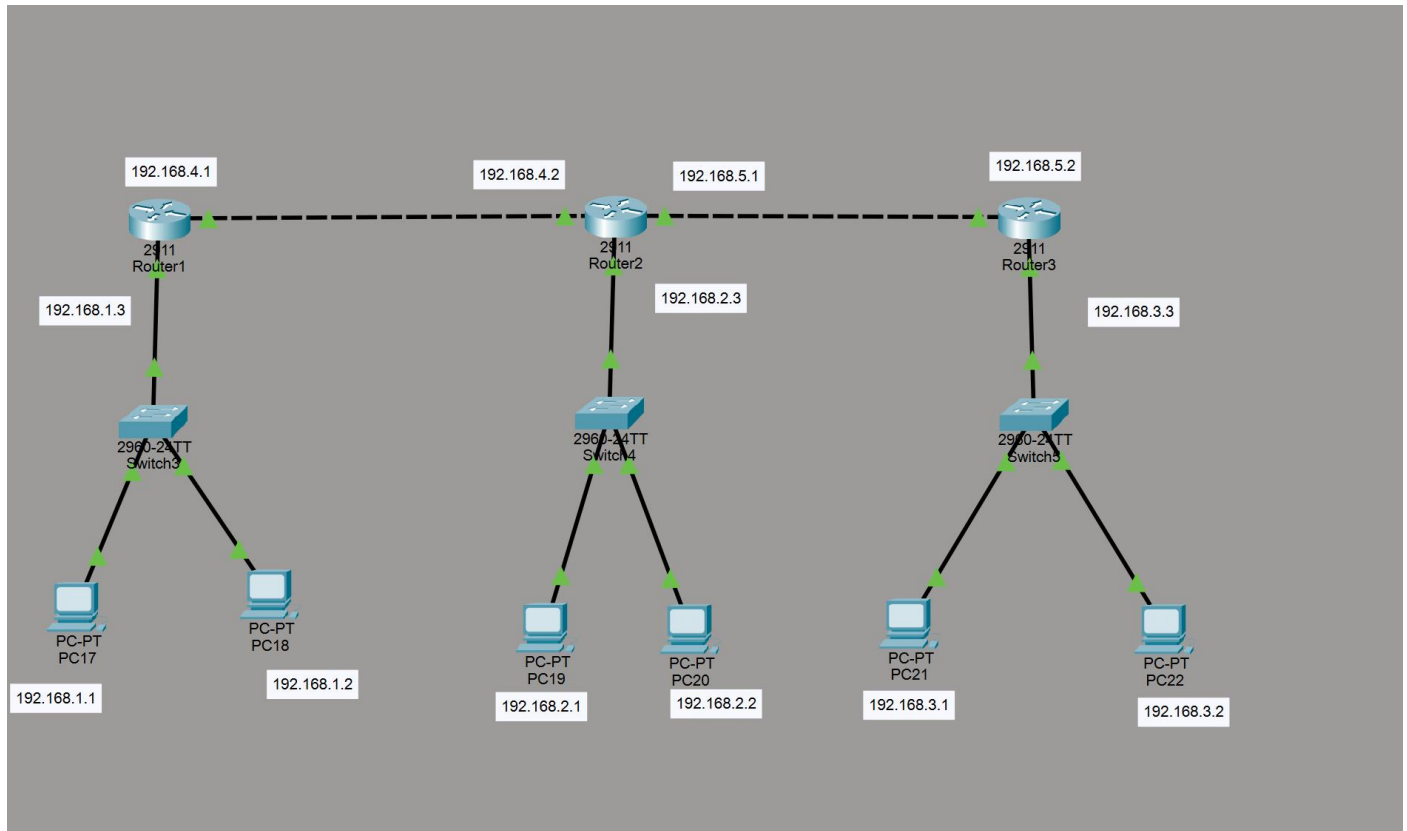
Input / Output Details

Devices Used

- Routers
- Switches
- PCs
- Copper Straight-Through Cable
- Serial Cable (if required)

Step-wise Output

Step 1 – Network Topology



Step 2 – IP Address Configuration

EXPERIMENT 4

Physical Config CLI Attributes

GLOBAL	GigabitEthernet0/0
Settings	Port Status ☒ On
Algorithm Settings	Link Speed ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto
ROUTING	Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto
Static	MAC Address 000C.857B.6801
RIP	IP Configuration
SWITCHING	IPv4 Address 192.168.1.3
VLAN Database	Subnet Mask 255.255.255.0
INTERFACE	Tx Ring Limit 10
GigabitEthernet0/0	
GigabitEthernet0/1	
GigabitEthernet0/2	

```
Router(config-router)#
Router(config-router)#end
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z
```

Step 3 – Static Route Configuration on Router 1

Router2

Physical Config CLI Attributes

GLOBAL	
Settings	
Algorithm Settings	
ROUTING	
Static	
RIP	
SWITCHING	
VLAN Database	
INTERFACE	
GigabitEthernet0/0	
GigabitEthernet0/1	
GigabitEthernet0/2	

Network Address

192.168.1.0/24 via 192.168.3.1

Add

Remove

```
Router(config)#ip route 192.168.3.0 255.255.255.0 192.168.3.1
Router(config)#
Router(config)#
Router(config)#
Router(config)#interface GigabitEthernet0/1
```

EXPERIMENT 4

Realtime Simulation										
Scenario 0	Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit Delete
New Delete Toggle PDU List Window	●	Successful	PC17	PC19	ICMP		0.000	N	0	(edit) (delete)

Scenario 0	Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit Delete
New Delete Toggle PDU List Window	●	Successful	PC21	Router3	ICMP		0.000	N	0	(edit) (delete)
	●	Failed	PC21	PC19	ICMP		0.990	N	1	(edit) (delete)
	●	Successful	PC19	PC21	ICMP		0.000	N	2	(edit) (delete)
	●	Successful	PC20	PC22	ICMP		0.000	N	3	(edit) (delete)
	●	Successful	PC22	PC19	ICMP		0.000	N	4	(edit) (delete)

Learning Outcome

After performing this experiment, I was able to:

- Understand the concept of routing and static routing.
- Configure IP addresses on routers and PCs.
- Configure static routes manually using Cisco IOS commands.
- Verify routing tables and network connectivity.
- Successfully establish communication between different networks using static routing.